



CLEAN IN ACTION
GREEN
FOR
APEC

INNOVATION AND APPLICATION OF NEW ENERGY POWER GENERATION TECHNOLOGIES

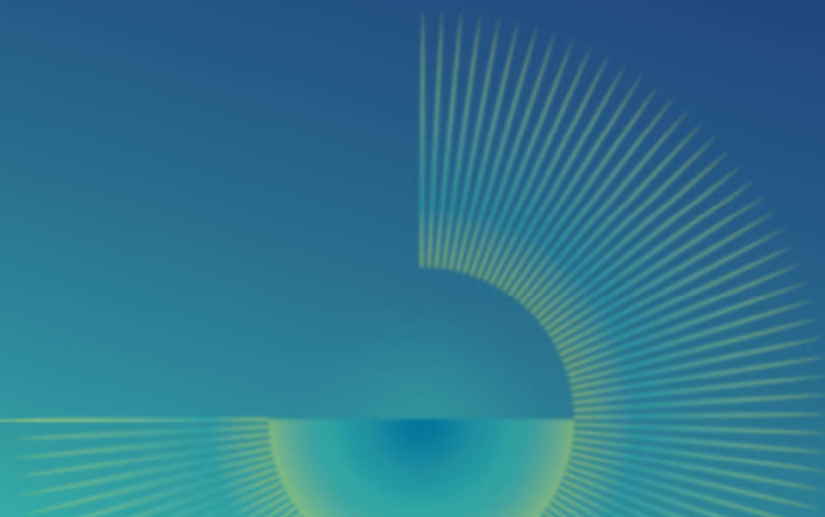
Speaker: Mina Guli

Date: 08.07.2026



ONE

TECHNOLOGY PATHWAYS FOR NEW ENERGY POWER GENERATION



1.1 DEVELOPMENT TRENDS OF NEW ENERGY POWER GENERATION TECHNOLOGIES

During the 14th Five-Year Plan period (2021–2025)

Installed capacity doubled: The total installed capacity of new energy reached **1.887 billion kW** (1,887 GW), comprising:

- ❑ Wind power: **640 million kW** (640 GW)
- ❑ Solar power: **1.2 billion kW** (1,200 GW)
- ❑ Biomass power: **47 million kW** (47 GW)

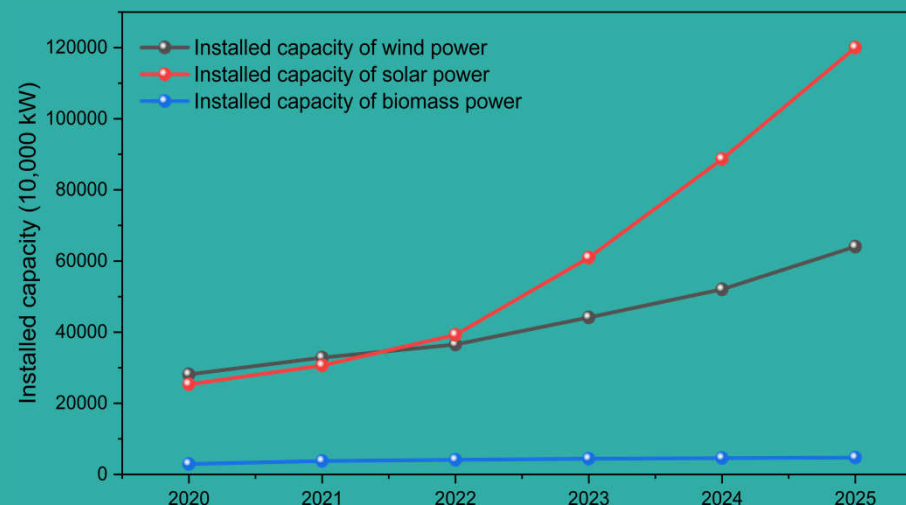
Green electricity consumption grew steadily: In 2025, the newly added renewable energy generation amounted to **519.3 billion kWh**, which already covered the increase in total electricity consumption of the whole society (516.1 billion kWh).



**Carbon peak
2030**



**Carbon neutrality
2060**



1.1 DEVELOPMENT TRENDS OF NEW ENERGY POWER GENERATION TECHNOLOGIES

At the start of the 15th Five-Year Plan period (2026–2030)

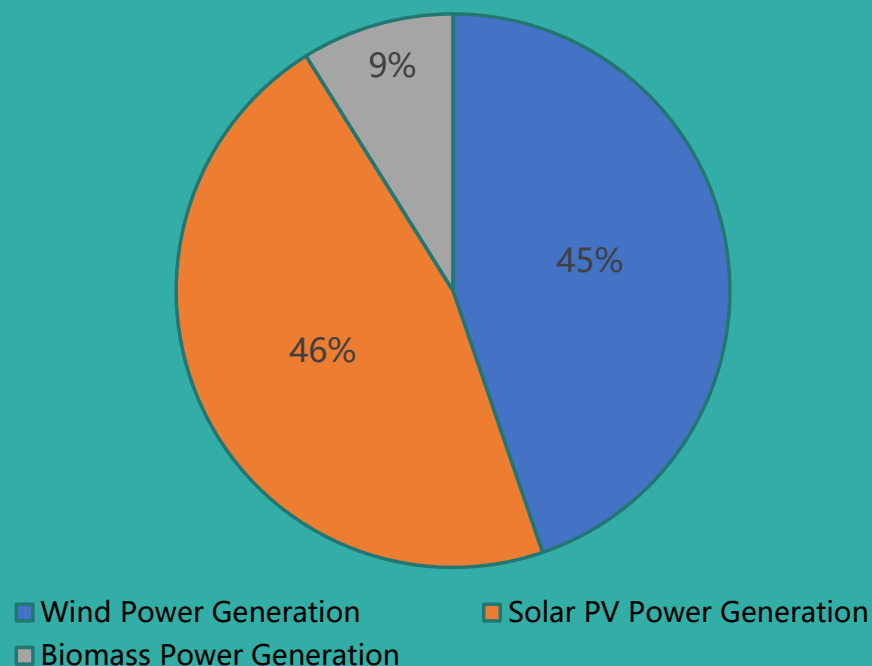
Building a new energy system: Develop major hydropower projects and integrated hydropower-wind-solar bases, as well as new energy bases in desert, Gobi, and barren lands, and offshore wind power bases.

Vigorously building a new-type power system: Construct smart grids, improve urban and rural distribution networks, scientifically deploy pumped-storage hydroelectricity, and vigorously develop new-type energy storage.

Basically completing a unified electricity market system: Expand cross-provincial and cross-regional electricity trading, improve ancillary service markets and capacity markets, and scale up green electricity trading and green certificate trading.



Share of New Energy Power Generation in 2025



Source: NEA, Renewable Energy Grid-Connected Operation in 2025

1.2 NEW ENERGY POWER GENERATION TECHNOLOGY ROUTES



Wind Power Generation

New installed capacity

2020
71.67 million kW



2025
120 million kW

- ❑ Significant economies of scale
- ❑ Good land compatibility
- ❑ No water consumption
- ❑ Fluctuating and intermittent
- ❑ High site requirements



Solar PV Power Generation

New installed capacity

2020
48.2 million kW



2025
318 million kW

- ❑ Abundant light resources
- ❑ Wide range of application scenarios
- ❑ Short construction cycle
- ❑ Highly intermittent
- ❑ Significantly affected by dust



Biomass Power Generation

New installed capacity

2020
5.43 million kW



2025
1.51 million kW

- ❑ Waste reuse
- ❑ Good stability
- ❑ Low fuel density
- ❑ Flue gas requires environmentally friendly treatment



1.3 DEVELOPMENT ROADMAP OF WIND POWER TECHNOLOGY

01 Last Century

Turbine: Fixed-pitch stall-regulated wind turbine.

Drivetrain/Gen: SCIG with gearbox.

Material: Mainly glass fiber reinforced; tower height < 40 m.



02 2000~2010

Turbine: Variable pitch technology.

Drivetrain/Gen: DFIG; early development of direct-drive synchronous generator.

Material: Mainly glass fiber reinforced composites; tapered pure steel tower.



03 2011~2020

Turbine: High towers, long flexible blades, & active yaw smart turbine technology.

Drivetrain/Gen: PM direct-drive & semi-direct-drive transmission.

Material: Carbon fiber; flexible high-steel towers over 120m & steel-concrete composite towers.



04 2021-Present

Turbine: Deep-sea floating offshore wind turbines;

Drivetrain/Gen: Medium-speed PM semi-direct-drive & direct-drive technology.

Material: Blade length exceeds 120m; large-scale application of carbon fiber pultrusion process.



1.3 INDUSTRIAL CHAIN OF WIND POWER TECHNOLOGY

Upstream: Raw Materials and Components

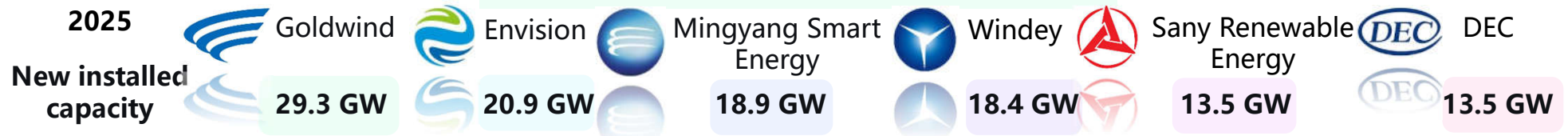
Basic Raw Materials



Key Components



Midstream: Turbine Manufacturing & Integration



Downstream: Wind Farm Development & Operation



Operation & Maintenance

Daily Inspections

Fault Diagnosis

Spare Parts Replacement

1.4 DEVELOPMENT ROADMAP OF PHOTOVOLTAIC TECHNOLOGY



Crystalline Silicon Technologies



PERC

Dielectric passivation layers (Al_2O_3 and SiN_x)



TOPCon

Ultra-thin tunnel oxide layer (SiO_2) and heavily doped poly-Si layer



HJT

Heterojunction of crystalline silicon (c-Si) and amorphous silicon

Back-Contact Branches

BC
IBC

TBC
THBC

HBC
HPBC
HIBC



Thin-Film Technologies



CdTe

- Low manufacturing cost
- Commercially mature



CIGS

- High theoretical efficiency
- Compatible with flexible fabrication



GaAs

- Highest conversion efficiency
- But relatively expensive



PSC

- High laboratory efficiency
- Short process flow

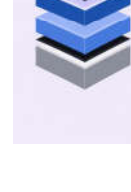


Tandem Technologies

Perovskite / Silicon Tandems



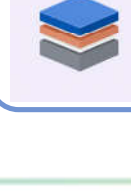
2-Terminal Tandem Solar Cells



3-Terminal Tandem Solar Cells



4-Terminal Tandem Solar Cells

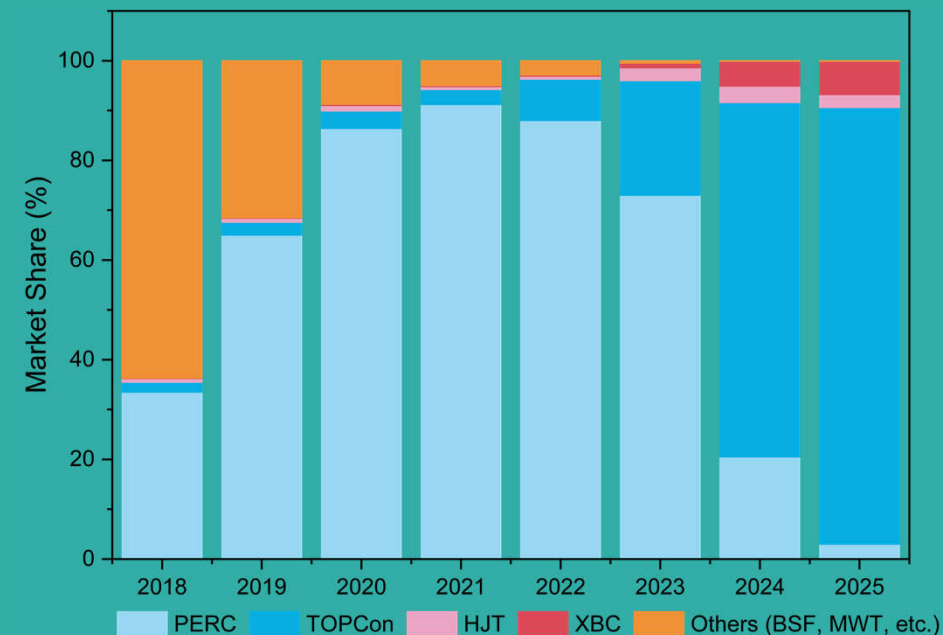
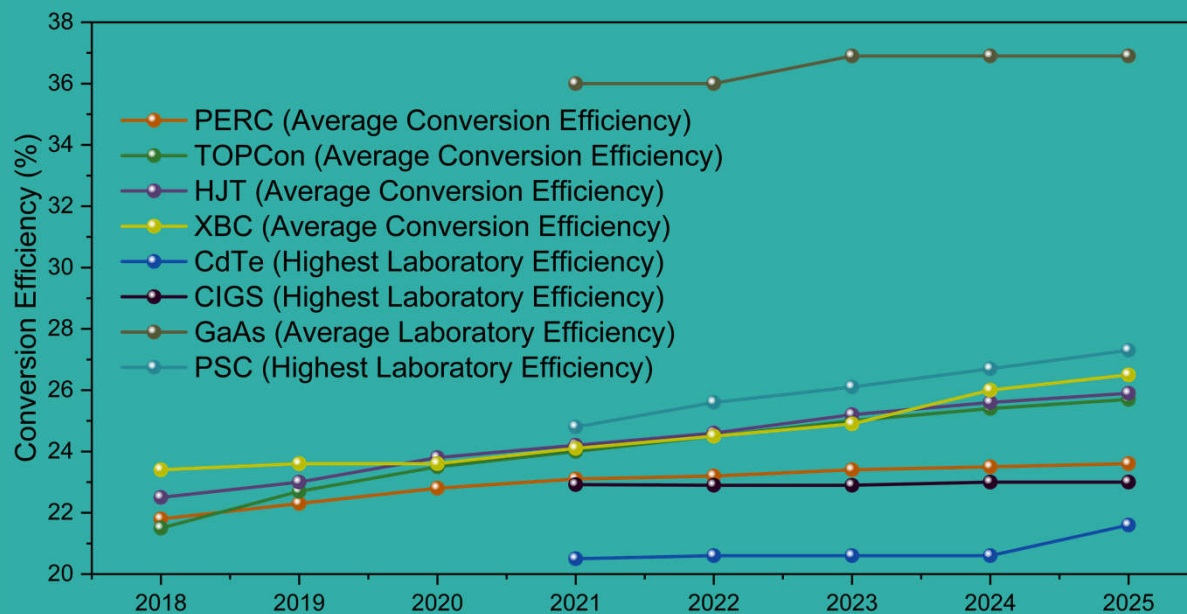


All-Perovskite Tandem Solar Cells



CIGS / Perovskite Tandem Solar Cells





- PERC is declining after peaking in market share, **with limited further efficiency gains**.
- TOPCon rapidly replaces PERC, reaching **87.6%** market share in 2025.
- TOPCon adoption **is cost-effective**, as most PERC line equipment can be reused.
- **n-type cells** remain the mainstream route for future crystalline-silicon efficiency gains.
- Perovskite and GaAs offer **high efficiency**, but commercialization paths differ greatly.

	2018	2025	Peak
PERC	33.5%	3%	2021 91.2%
TOPCon	2%	87.6%	2025 87.6%
HJT	0.7%	0.6%	2024 3.3%
XBC	0.1%	6.7%	2025 6.7%



Perovskite Industrialization Development

The production of rigid perovskite solar cells in the industry has entered the mass-production stage.

Ten pilot production lines of 100 MW or above have been put into operation, and **three GW-scale production lines** are under construction.

By the end of **2025**, **nine MW-scale** ground demonstration power stations using single-junction perovskite modules have been built, with the largest single project reaching an installed capacity of **8.6 MW**.



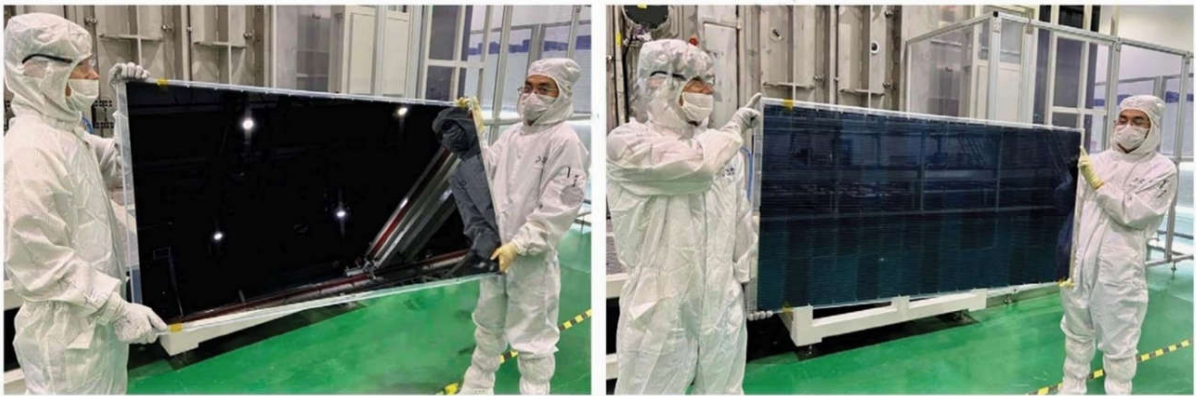
Highest Record Comparison

Types of photovoltaic cells	Highest efficiency	Record-holding Institution
Passivated Emitter and Rear Cell (PERC)	24.06%	LONGi
Tunnel Oxide Passivated Contact (TOPCon)	27.02%	Jinko Solar
Heterojunction with Intrinsic Thin-layer (HJT)	27.08%	Trina Solar
Interdigitated Back Contact (IBC)	24.3%	Aiko Solar
TOPCon Back Contact/TOPCon Hybrid Back Contact (TBC/THBC)	28.13%	Trina Solar
Heterojunction Back Contact (HBC)	28.2%	JA Solar & Goldstone Energy
Hybrid Passivated Back Contact/Hybrid Interdigitated Back Contact (HPBC/HIBC)	28.13%	LONGi
Cadmium Telluride (CdTe)	20.61%	Longyan Energy
Copper Indium Gallium Selenide (CIGS)	23.0%	Hanergy/ Solibro
Gallium Arsenide (GaAs)	37.5%	Derong Technology Co., Ltd.
Perovskite Solar Cells (PSC)	28.0%	Hainan University
Perovskite/Crystalline Silicon Tandem Solar Cell (PSC/Si tandem)	35.2%	LONGi

1.4 INDUSTRIALIZATION TRENDS OF PHOTOVOLTAIC TECHNOLOGIES

Perovskite / Crystalline Silicon Tandem Industrialization Development

Perovskite / Crystalline Silicon Tandem Conversion Efficiency	2021	2022	2023	2024	2025
Highest laboratory efficiency (%)	25.7	29.55	33.9	34.6	34.85
Highest module efficiency (%)	-	-	26.34	26.36	30.6



GCL Optoelectronics 2024 Perovskite Industrialization Progress Report



2021-2022

Technology R&D Stage

Focused on distributed applications of mass-produced perovskite modules; no commercial tandem module efficiency had been released

2023-2024

Large-area Pilot Production Stage

GCL made continuous industrialization progress and planned its first GW-scale tandem module base in Kunshan, Jiangsu, scheduled for completion in 2026

2025

Technology Breakthrough Stage

Trina Solar advanced low-shading conductive, adaptive heterojunction, and photon management technologies, improving both efficiency and power output

1.5 DEVELOPMENT ROADMAP OF BIOMASS POWER GENERATION TECHNOLOGY

Scale-up Stage

January 1, 2006 《Renewable Energy Law of the People's Republic of China》
January 20, 2020 《Several Opinions on Promoting the Healthy Development of Non-hydro Renewable Energy Power Generation》

- ❑ Benchmark on-grid tariff
- ❑ Full guaranteed purchase by the power grid
- ❑ Subsidies for new projects based on revenue availability

Cogeneration Stage

June 1, 2022
《14th Five-Year Plan for Renewable Energy Development》

- ❑ Combined heat and power
- ❑ Multi-source solid waste co-processing

Carbon-negative Stage

《Blue Paper on the Development Potential of Zero-carbon Biomass Energy under the 3060 Goals》

- ❑ Declining competitiveness of power generation
- ❑ Sustainable aviation biofuel, biomethane, green methanol



1.5 BIOMASS TECH ROADMAP

Waste-to-energy Incineration Power Generation



Cost reduction and
efficiency improvement

ACC

Agricultural and Forestry Biomass Power Generation



Full-scale combined heat and power

Biogas Power Generation



ecological circular cogeneration

Medium Term



Multi-source waste co-processing



Fuel and equipment upgrading



High-value utilization

Long Term



Fly ash vitrification
CCUS carbon injection wells



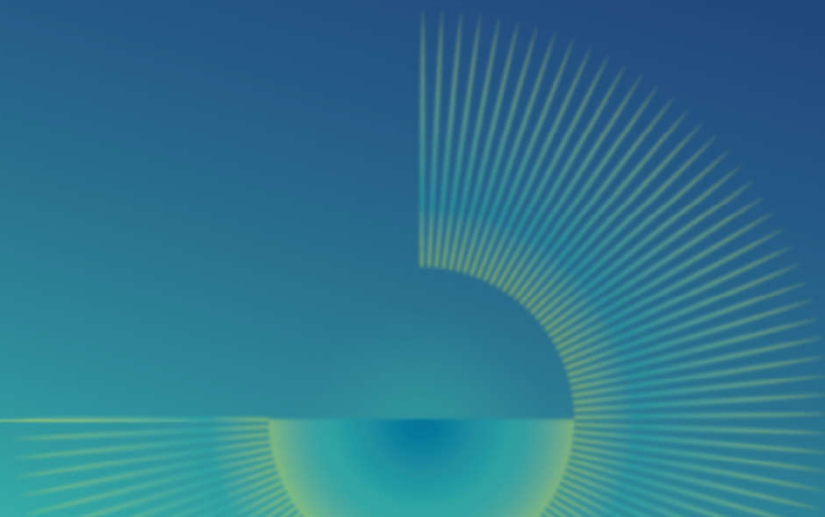
Multi-energy complementarity,
microgrid, hydrogen production



Deep power grid integration

TWO

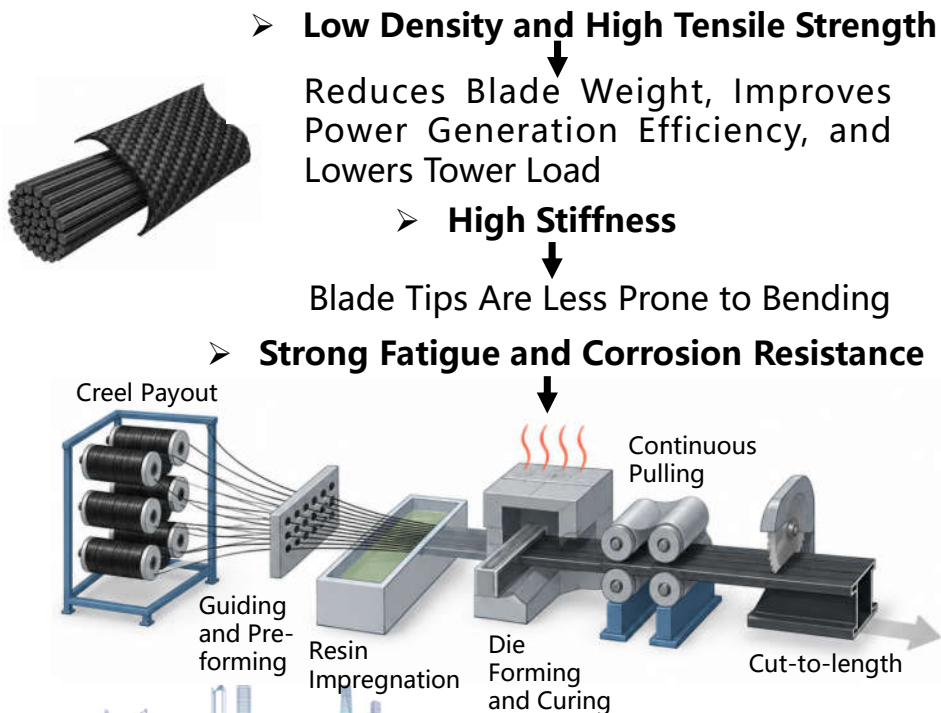
BREAKTHROUGH INNOVATIONS IN NEW ENERGY POWER GENERATION



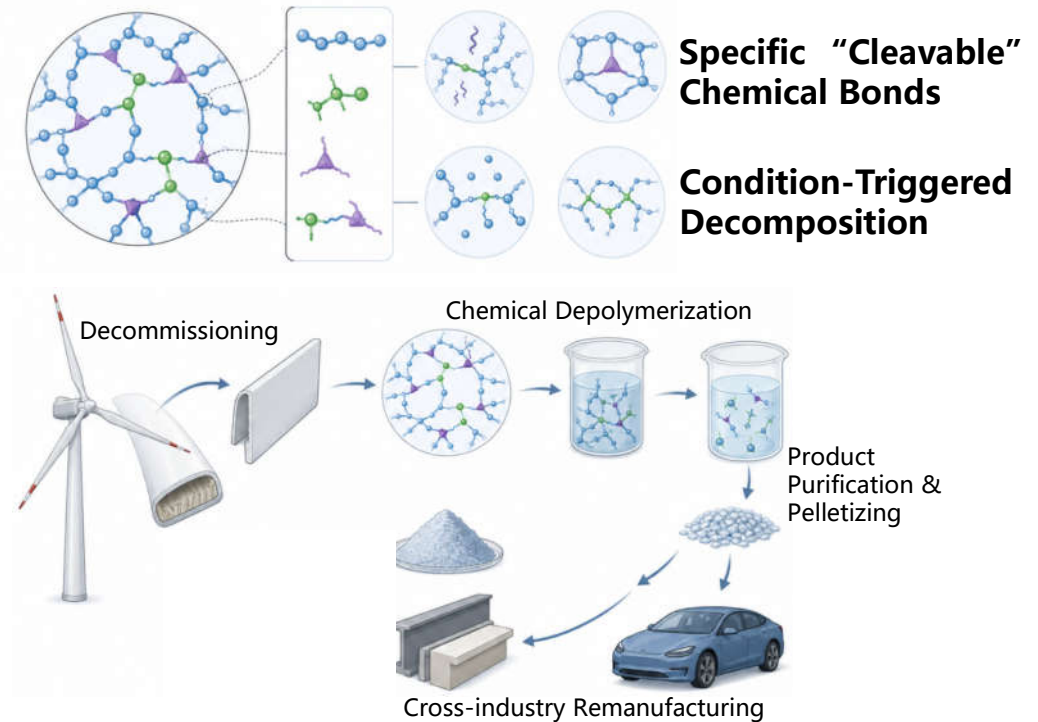
2.1 KEY INNOVATIVE TECHNOLOGIES IN WIND POWER GENERATION (OFFSHORE WIND POWER)

Blade Materials

(1) Carbon Fiber Pultrusion Technology



(2) Recyclable Thermosetting Epoxy Resin



2.1 KEY INNOVATIVE TECHNOLOGIES IN WIND POWER GENERATION (OFFSHORE WIND POWER)

Drivetrain and Power Generation Technologies

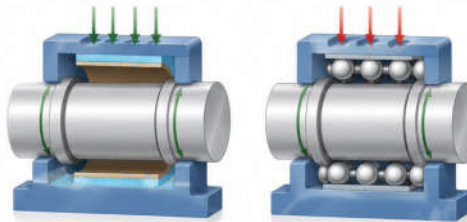
(1) Medium-Speed Semi-Direct Drive Technology



Medium-speed gearbox Medium-speed permanent magnet synchronous generator

- Balances generator size, weight, and cost
- Lower drivetrain failures and O&M costs
- High efficiency across wide wind speeds
- Strong LVRT/HVRT and grid support

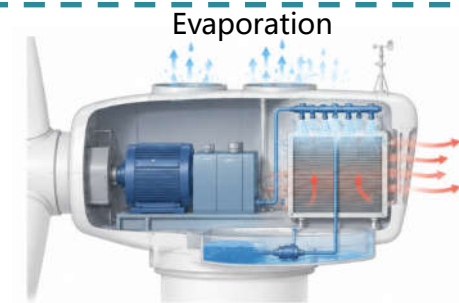
(2) Sliding Bearing Technology



Sliding bearing Rolling Bearing

- High load-bearing capacity
- Lower nacelle weight and foundation costs
- Good vibration damping
- No fatigue life limit

(3) Evaporative Cooling Technology



- High cooling efficiency
- Phase-change cooling lowers auxiliary power use
- Stable and uniform temperature
- Closed-loop protection against marine corrosion

2.1 KEY INNOVATIVE TECHNOLOGIES IN WIND POWER GENERATION (OFFSHORE WIND POWER)

Semi-submersible Floating Platform

- ❑ High maturity
- ❑ Strong stability
- ❑ Large-MW turbines

Tension Leg Platform

- ❑ Small motion
- ❑ Compact footprint
- ❑ Ultra-large-MW turbines

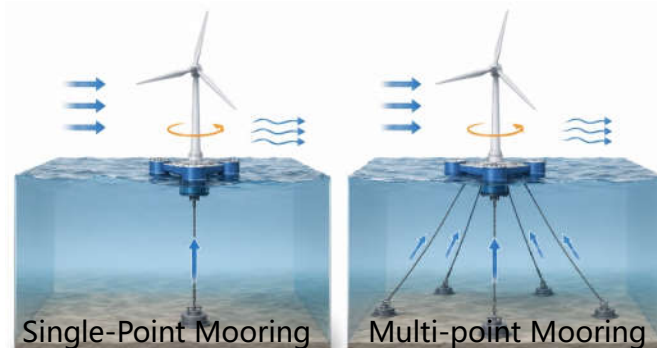


Single-Point Mooring

- ❑ Lower resistance
- ❑ Self-alignment
- ❑ Better multi-rotor inflow

Multi-Point Mooring

- ❑ Distributed loads
- ❑ Good redundancy
- ❑ Stable orientation



High-Voltage Dynamic Subsea Cable Technology



Lazy-wave configuration

Motion-adaptive design



Steep-wave configuration

Lower failure risk



Free-hanging configuration

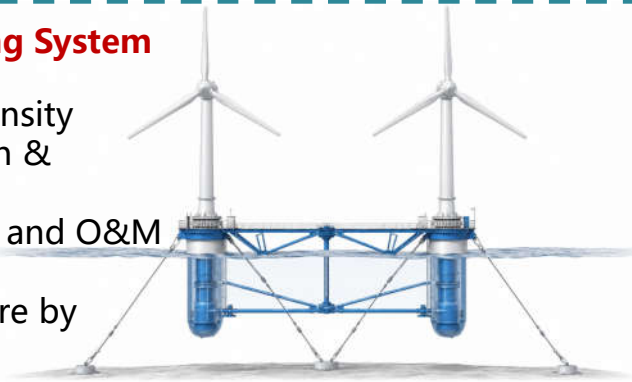
Deep-sea, far-offshore & large-capacity applications



2.1 KEY INNOVATIVE TECHNOLOGIES IN WIND POWER GENERATION (OFFSHORE WIND POWER)

Twin-turbine Floating System

- ❑ Higher turbine density
- ❑ Shared foundation & mooring
- ❑ Lower installation and O&M cost
- ❑ Better wind capture by weathervaning



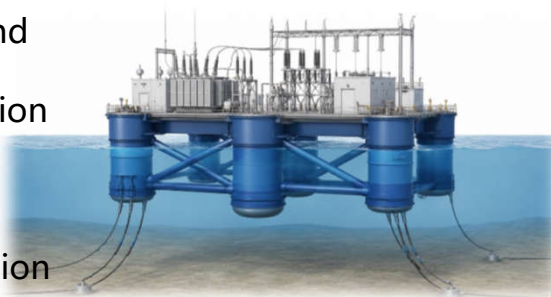
Wind-Fishery Integrated Floating Platform

- ❑ Shared sea-space utilization
- ❑ Wind power + marine aquaculture
- ❑ Embedded aquaculture cage
- ❑ Smart unmanned monitoring



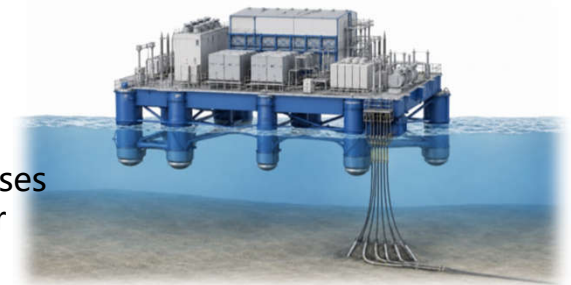
Floating Offshore Substation

- ❑ Supports deep-sea wind farms
- ❑ Reduces fixed-foundation works
- ❑ Enables centralized planning
- ❑ Offshore power collection hub



Floating HVDC Converter Platform

- ❑ Long-distance power export
- ❑ High-capacity transmission
- ❑ Supports GW-scale bases
- ❑ Reduces cable corridor pressure



2.1 OFFSHORE WIND PROJECTS IN PRC

Three Gorges Lead

Grid-connected on 2021.12.7

PRC's first floating offshore wind platform

- ❑ Hub height: ~107 m
- ❑ Rotor diameter: 158 m
- ❑ Displacement: ~13,000 ton

CNOOC Guanlan

Grid-connected: 2023.05

PRC's 1st deep-sea floating platform

- ❑ Offshore: 136km Rotor: 158m
- ❑ Depth: 120m Cable: 5km
- ❑ Capacity: 7.25 MW
- ❑ Annual Output: 22million kWh

CHN Energy Sharing Platform

Operation: 2024.06.28

PRC's 1st floating wind-aquaculture platform

- ❑ Offshore: ~30 km; Depth: ~35 m
- ❑ Capacity: 4 MW;
- ❑ Rotor diameter: 130 m
- ❑ Fish cage: ~10,000 m³ / 50,000 fish

Three Gorges Pilot

Installed: 2026.05

PRC's largest single-turbine platform

- ❑ Offshore: >70km
- ❑ Depth: >50m
- ❑ Displacement: 24,100ton
- ❑ Rotor: 252m Area: ~50,000m²
- ❑ Capacity: 16 MW
- ❑ Est. annual Output: 44.65million kWh



2.2 KEY INNOVATIVE TECHNOLOGIES IN PV POWER GENERATION

Offshore PV

Pile-Fixed Offshore System

- ❑ High maturity & wind/wave resistance
- ❑ Synergizes with aquaculture/salt pans

Pontoon-Based Floating PV System

- ❑ Minimal seabed disturbance
- ❑ Broader water depth range

Semi-Submersible Floating PV Platform

- ❑ High stability & wind/wave resistance
- ❑ Supports tilted modules & equipment

Wave-Resistant Non-Overtopping Floating PV Platform

- ❑ Less seawater contact
- ❑ Lower corrosion & leakage risks

Flexible Membrane-Buoyancy Ring Floating PV Platform

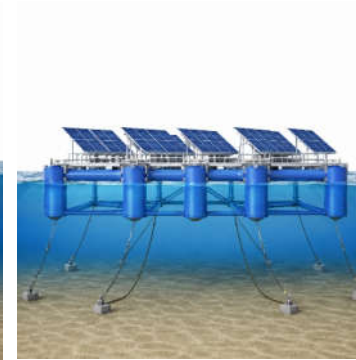
- ❑ Lightweight & minimal material usage
- ❑ Reduced wave impact & mooring loads

Tensioned Floating Island PV Platform

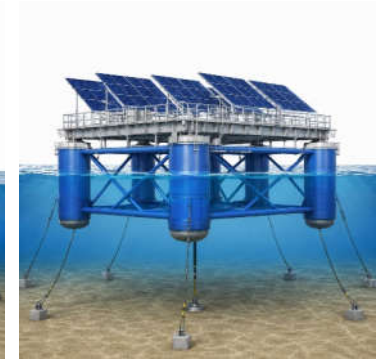
- ❑ Highly adaptable & modular
- ❑ Eco-friendly composite materials



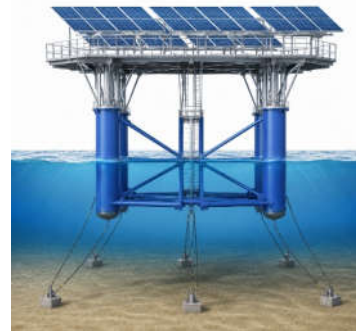
Pile-Fixed Offshore System



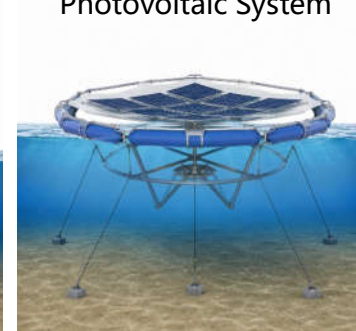
Pontoon-Based Floating Photovoltaic System



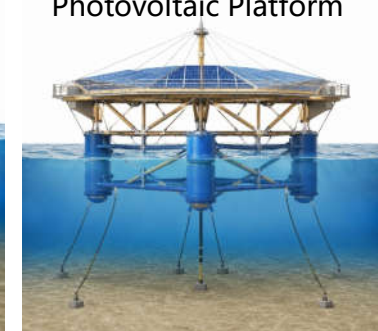
Semi-Submersible Floating Photovoltaic Platform



Wave-Resistant Non-Overtopping Floating PV Platform



Flexible Membrane-Buoyancy Ring Floating PV Platform



Tensioned Floating Island Photovoltaic Platform

2.2 KEY INNOVATIVE TECHNOLOGIES IN PV POWER GENERATION

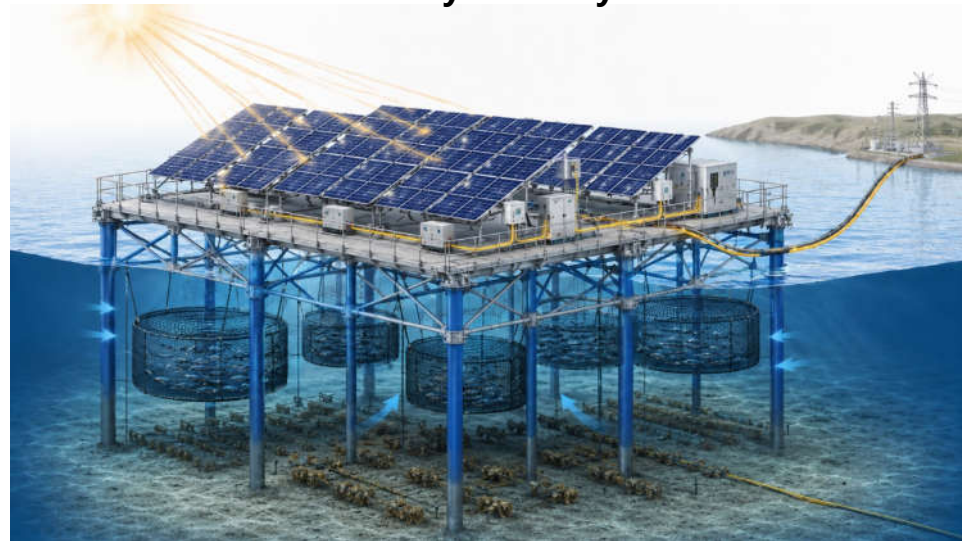
Offshore PV

Bifacial Module



- ❑ Unobstructed surface & high reflection
- ❑ Higher energy yield per area

Fishery-Solar Hybrid



- ❑ For mudflats, aquaculture & shallow seas
- ❑ Higher land & water utilization
- ❑ Increased economic benefits
- ❑ Lower aquaculture energy costs



2.2 OFFSHORE PV PROJECTS IN PRC

1 Hengzhou Tiantangling Agrivoltaic-Storage



Jul. 2023

Installed 488 MW | 494 GWh/y

First Batch On-grid

Agriculture

Complementary energy storage
51 MW/102 MWh

2 Longyuan Tianjin Haijing Salt-PV Project



Sep. 2023

On-grid 580 MW | 770 GWh/y

Salt-PV-Fishery

equipped with
500 kV substation

Achieve One Site, Three Uses

3 Zhoushan Dengbu Island Fishery-PV Project



Jul. 2024
Phase I

213 MWp planned

Dual 110 kV Feed

Fishery-PV

Smart Aquaculture

4 Rudong PV-H₂-Storage Project



Apr. 2025

400 MW | 468 GWh/y

PV-H₂-Storage

1,500 Nm³/h H₂

Complementary energy storage
60 MW/120 MWh

5 Dongshan Xingchen Offshore PV



Apr. 2025

180 MW | 300 GWh/y

High-wind Offshore

48 m/s

Complementary energy storage
18 MW/36 MWh

6 Yantai Zhaoyuan HG30 Offshore PV



May 2025

AC 400 MW | DC 539 MWp

Fixed-pile PV

Bifacial Modules

2-6.2 km Offshore

7 Qingdao Jimo Offshore PV



May 2025

1.15 GW planned

Fishery-PV

Pile-fixed PV

Underwater planning and
development aquaculture

8 Qingdao Refinery Floating PV



Jul. 2025
Put into use

7.5 MW | 16.7 GWh/y

Rise and fall in
sync with tide

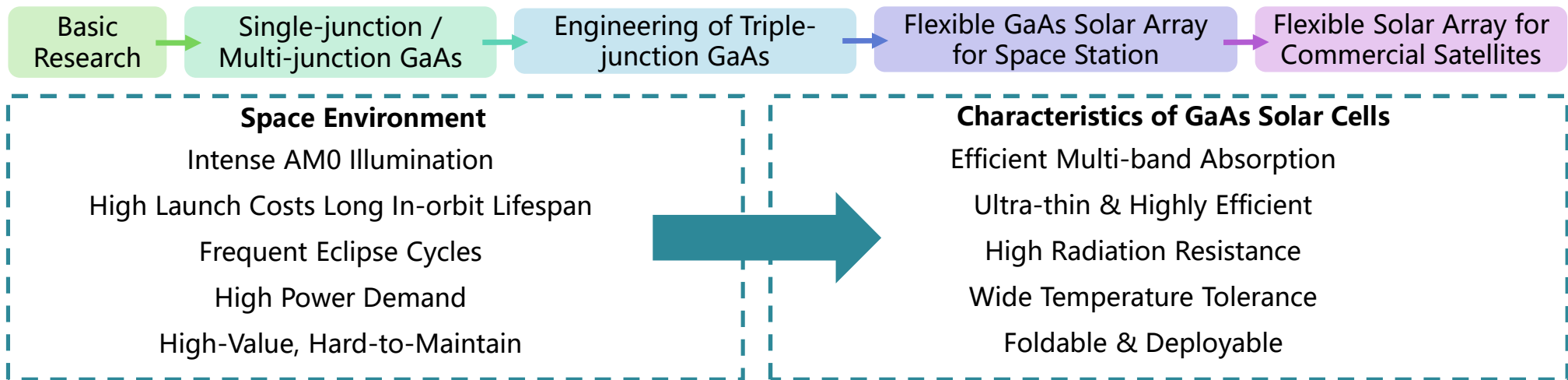
Link with Green H₂
demonstration park

First industrial operation floating
PV project in a fully seawater



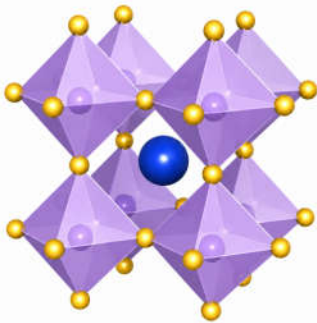
2.2 KEY INNOVATIVE TECHNOLOGIES IN PV POWER GENERATION (**SPACE PV**)

Flexible Gallium Arsenide (GaAs) Solar Array



2.2 KEY INNOVATIVE TECHNOLOGIES IN PV POWER GENERATION (SPACE PV)

Perovskite solar cells



- Strong light absorption
- Long carrier diffusion length
- High defect tolerance
- Low-temperature fabrication
- Tunable bandgap via composition

- Poor Stability
(Sensitive to moisture, heat, light & bias)
- Low Large-Area Efficiency

Compositional Engineering

Interface Passivation

Additive Engineering

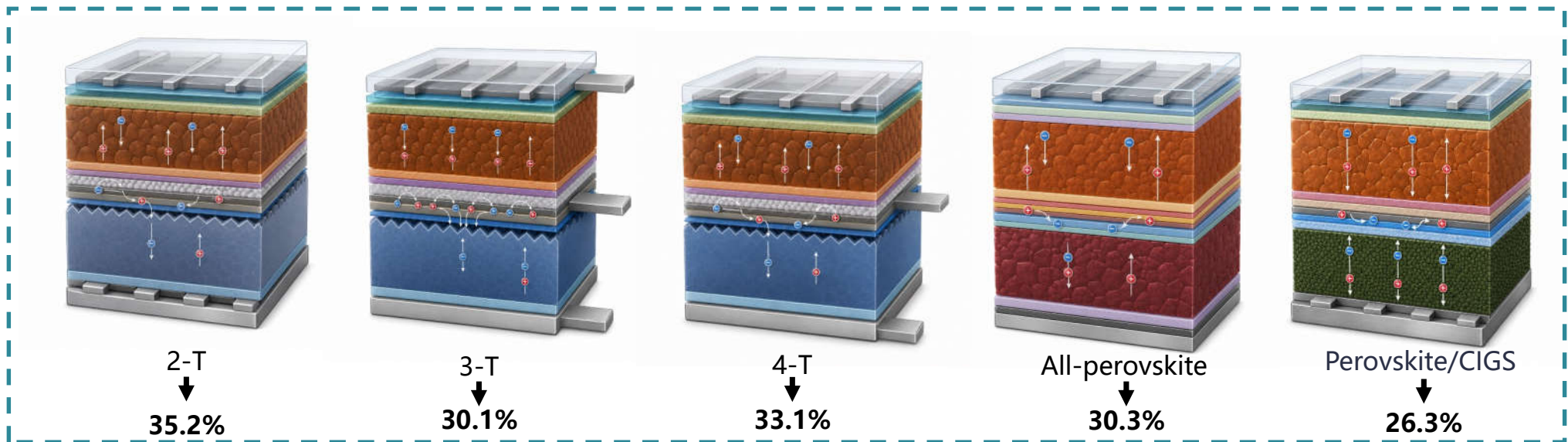
Large-Area Fabrication

Low-Dimensional/3D Composite Structures

Time	Project/Satellite	Institutions/Companies
2023.12	Honghu-2 Satellite	GCL Perovskite / LandSpace / Hongqing Technology
2024.5	Guangyin Technology Space Experiment	Guangyin Technology
2024.5.31	Fudan Information Satellite	Shanghai Fuxi Xinkong
2024.8.29	Suxing-1 Satellite	Shanghai Fuxi Xinkong
2024.11.11	Tianyan-24 Satellite	Shanghai Fuxi Xinkong
2026.5.24	Shenzhou-23 (PSC/Si tandem)	CAS (Space Utilization Center, Institute of Semiconductors, etc.)

2.2 KEY INNOVATIVE TECHNOLOGIES IN PV POWER GENERATION (**SPACE PV**)

Perovskite / Tandem Cells



- Wide-bandgap top cell: short-wavelength / high-energy photons
- Narrow-bandgap bottom cell: long-wavelength / low-energy photons
- Beyond the single-junction Shockley–Queisser limit
- Higher efficiency ceiling
- Compatible with silicon PV industry
- Strong potential for space and flexible applications

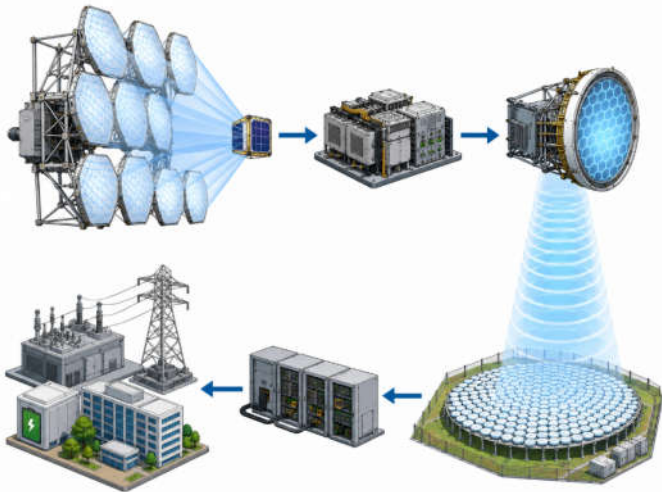
Shenzhou-23 Perovskite Solar-Cell Experiment

Real-space extreme-environment testing

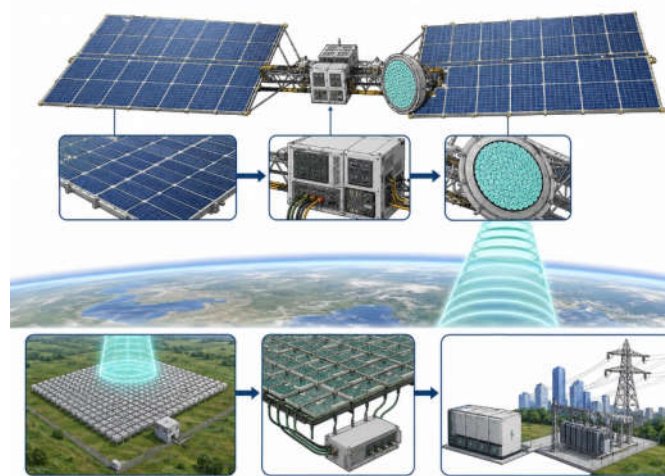
Efficiency degradation data collection

2.2 KEY INNOVATIVE TECHNOLOGIES IN PV POWER GENERATION (**SPACE PV**)

Space Solar Power Station & Wireless Power Transmission



Concentrating Type



Non-Concentrating Type

- Space solar collection & PV conversion
- Microwave wireless power transmission
- Beam steering & phased-array control
- Rectenna reception, rectification & grid integration

2022

2023

2026

2030

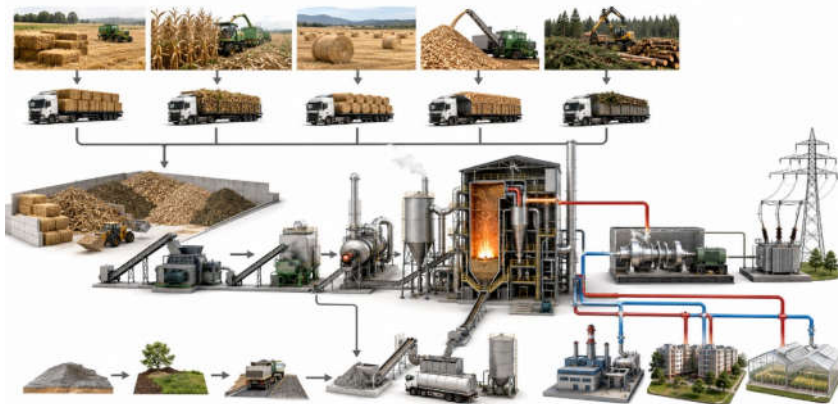
"Zhuri Project" SSPS ground verification system

OMEGA-SSPS ground test system; OMEGA 2.0 proposal

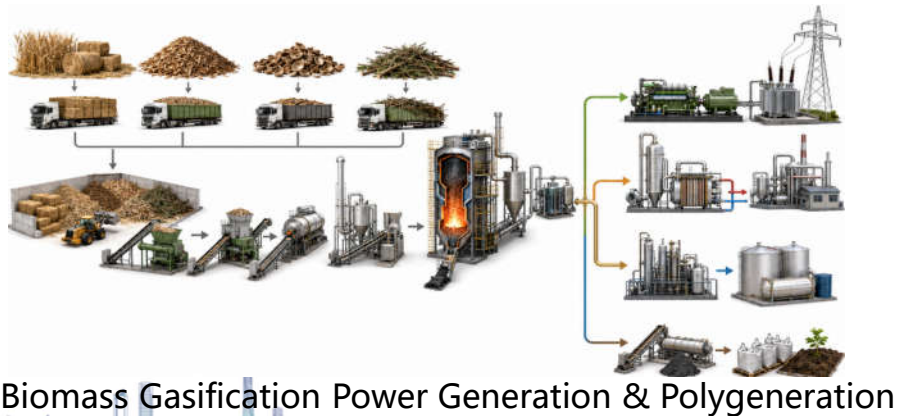
Distributed OMEGA SSPS design; one-to-multiple moving-target microwave power beaming

Megawatt-scale in-orbit test

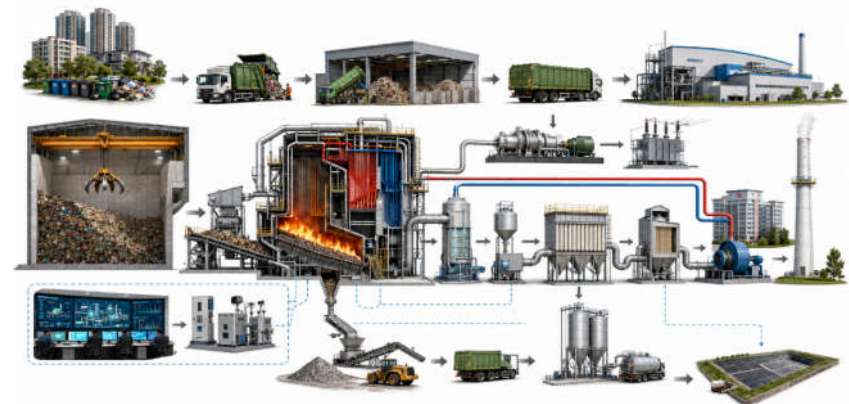
2.3 KEY INNOVATIVE TECHNOLOGIES IN **BIOMASS** POWER GENERATION



Agroforestry Biomass Direct Combustion & CHP



Biomass Gasification Power Generation & Polygeneration



Municipal Solid Waste Incineration Power Generation



Biogas / Biomethane Power Generation & CHP

2.4 AI-DRIVEN RENEWABLE POWER FORECASTING

1. Data Input

Data Type	Data Content
Weather Data	Irradiance, wind speed, wind direction, temperature, humidity, cloud cover
Numerical Weather Prediction NWP	15-min to 7-day weather forecasts
Plant Operation Data	SCADA, inverter data, turbine status, historical power
Spatial & Environmental Data	Terrain, elevation, shading, satellite cloud maps
Grid Operation Data	Curtailement records, grid-connection voltage, dispatch instruction
Market & Load Data	Electricity prices, load curves, imbalance assessment rules

2. Data Fusion & Cleaning

Data Cleaning
Anomaly Detection
Missing Data Imputation
Time Synchronization
Feature Engineering



Data Support Operation Features

3. AI-Ready Features

Time Features

Hour

Holiday

Season

Sunrise / Sunset

Weather Features

Wind Speed

Cloud Cover

Irradiance

Temperature & Humidity

Operation Features

Historical Power

Equipment Status

Curtailement Status

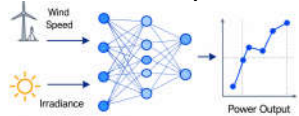
2.4 AI-DRIVEN RENEWABLE POWER FORECASTING

AI prediction model

Traditional ML & Shallow Models

ANN/BP

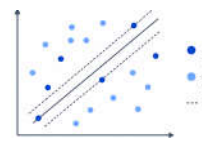
Nonlinear mapping of wind speed-power & irradiation-power



Simple and easy to deploy

SVM/SVR

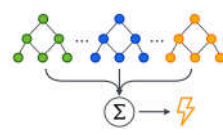
Small sample wind & solar power prediction



Stable under small data

Random Forest

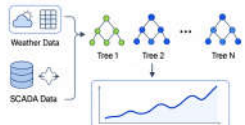
Power prediction anomaly identification



Strong noise resistance

XGBoost / LightGBM

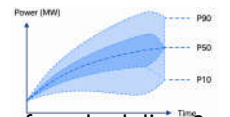
Tabular meteorological data and SCADA data prediction



High precision & good interpretability

Quantile Regression

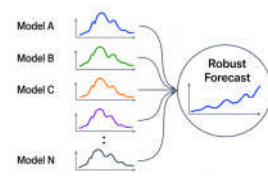
Probabilistic forecast Prediction of the P10/P50/P90 range



for scheduling & market transactions

Ensemble Learning

Combine diverse models

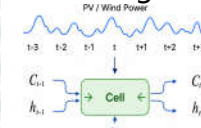


Improve robustness

Deep Learning for Temporal & Spatial Features

LSTM / GRU

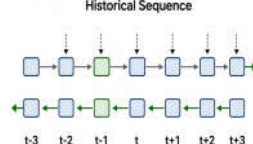
Ultra-short-term & short-term wind & pv forecasting



handle time series data

BiLSTM

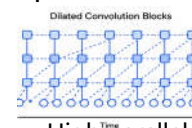
Historical Sequence Feature Extraction



Stronger expression ability

TCN

Long-term sequence prediction



High parallel efficiency

CNN

Satellite cloud images, local meteorological fields



Extract spatial features

ConvLSTM

Cloud-based prediction of photovoltaic power generation



study the changes in space & time

Transformer

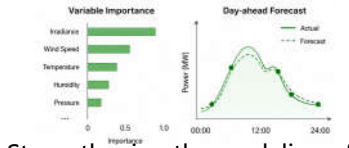
Multivariate & multi-step power prediction



Strong modeling capability for long sequences

TFT / iTransformer

multiple meteorological variables pv prediction

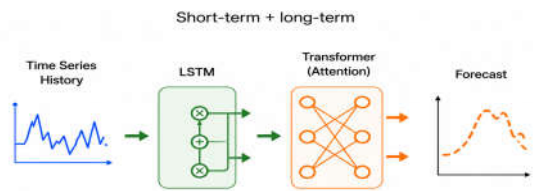


Strengthening the modeling of relationships between variables

Hybrid Models

LSTM + Transformer

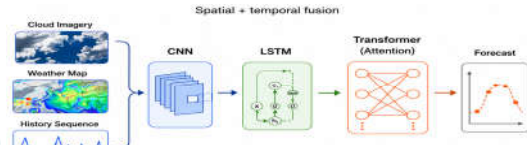
Short-term/Day-ahead Wind/PV Forecast



Short-term memory + Long sequence attention

CNN-LSTM-Transformer

Cloud chart, meteorology, historical power fusion prediction



Space + Time + Attention Integration

GNN / ST-GNN

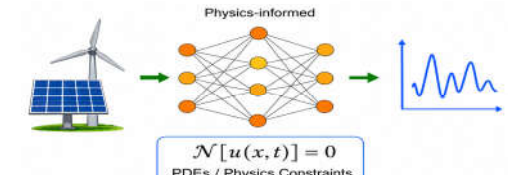
Multi-station & regional renewable energy forecasting



Modeling spatial correlation

PINN

PV/Wind Physical Integration Prediction



Improve credibility and generalization ability

2.4 AI-DRIVEN RENEWABLE POWER FORECASTING

Engineering application



Renewable Energy Grid
Integration and Dispatch



Energy Storage Optimization and Control



Virtual Power Plant (VPP) Aggregation

AI Power Forecasting Results



Microgrid Dynamic Response



Power / Electricity Market Trading



Intelligent O&M

2.4 PRC AI FORECASTING PROJECTS

1 CSG “Kuaifu” Forecasting System



High-res NWP + AI

Service 600+ renewable plants
Wind/PV Forecasting
Grid Dispatch

2 State Grid Xinjiang AI Forecasting System



Multi-algorithm AI forecasting

forecast accuracy:
Wind 96% · PV~98%
Big Data · Renewable Absorption

3 State Grid Fujian Distributed PV Model



Grid-based ML prediction

County-level distributed PV
Distribution Grid
Lightweight ML

4 Huadian–Huawei–Jiutian AI Weather–Power Model



Pangu Weather + Edge AI

15-min accuracy 97.24%
Weather Model
Edge Forecasting

5 Shenzhen Energy–Huawei Cloud Forecasting System



Pangu-based power forecast

Prediction accuracy improve:
Wind +10% · PV +2%
Cloud AI · Multi-energy Forecast

6 State Power Rixin “Kuangming” Model



Dynamic GNN + Attention

Power & market forecasting
GNN
Extreme Weather

7 Goldwind /PV Forecasting Technology



Weather Big Model + AI

Prediction accuracy +3%–5%
Wind/PV Forecast
Big Model

8 CHN Energy “Qingyuan” Power Generation Model



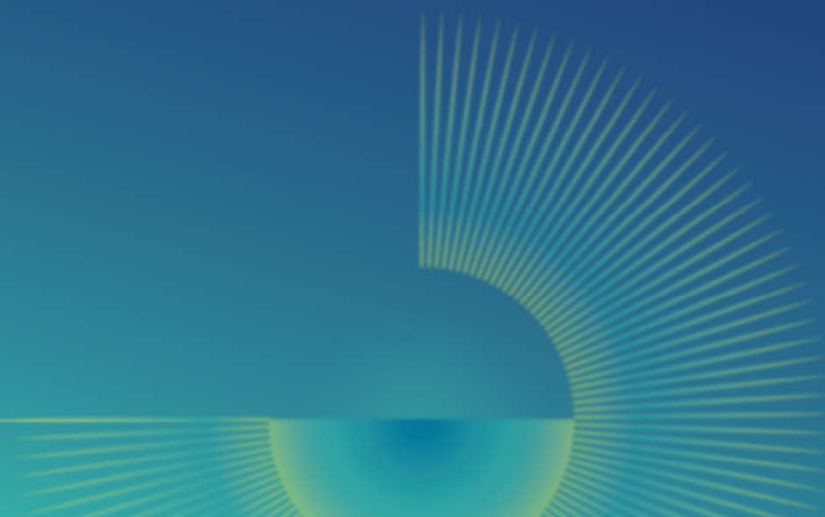
Power-sector foundation model

Forecast accuracy improve+2.8%
Large Model
Trading Decision

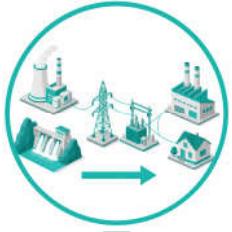


THREE

NEW POWER SYSTEM INTEGRATION & APPLICATION



NEW POWER SYSTEM INTEGRATION & APPLICATION



Traditional Power System

- ❑ Dominated by controllable generation (thermal, hydro)
- ❑ Generation follows load variations
- ❑ One-way generation-side regulation



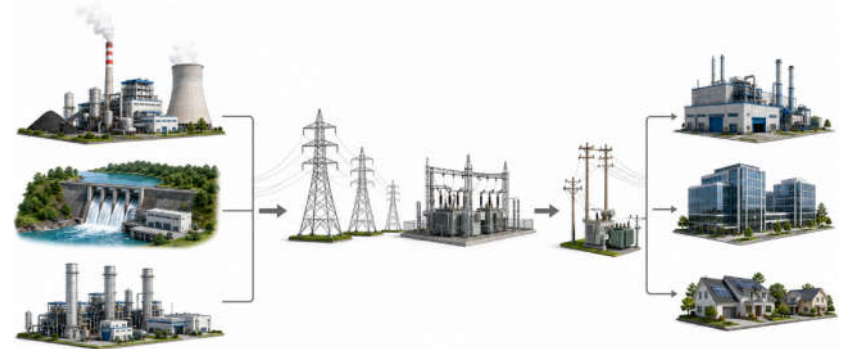
Challenges with Renewable Integration

- ❑ High penetration of wind and solar power
- ❑ Increasing randomness, volatility, and intermittency
- ❑ Limited flexibility for renewable integration

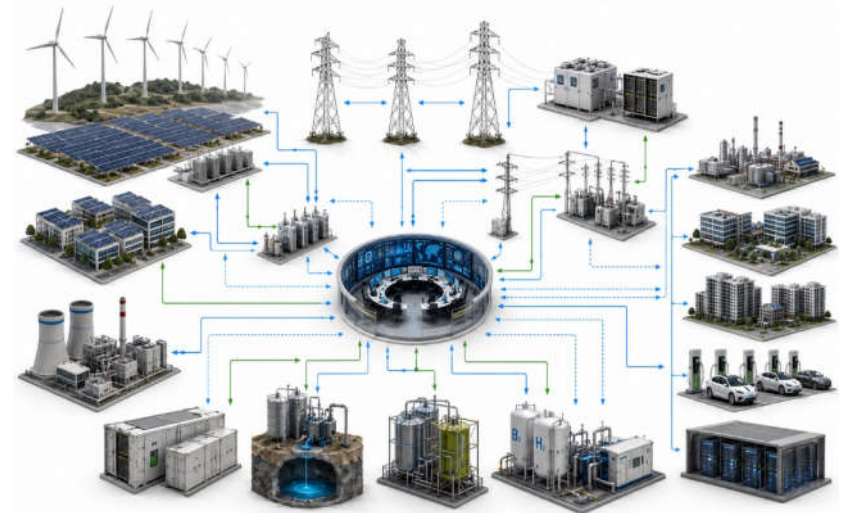


New Power System

- ❑ High renewable energy penetration
- ❑ Multi-side coordinated flexibility management
- ❑ Interactive source-grid-load-storage ecosystem



"Generation Follows Load"



"Source-Grid-Load-Storage Integration"

NEW POWER SYSTEM INTEGRATION & APPLICATION

Source-Grid-Load-Storage Integration

- Coordinated planning and control of generation, grid, loads and storage for efficient renewable use and real-time power balance.

Key Features

- Source Follows Load
- Load Responds to Supply
- Storage Balances Fluctuations
- Grid-Friendly Operation
- Smart Coordinated Control

Key Benefits

More Renewables

Lower Costs

Greater Flexibility

Reliable Supply

"Source"



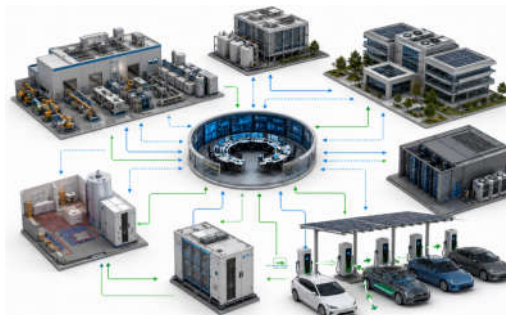
New energy prediction, multi-energy complementarity, active support

"Grid"



Intelligent dispatching, flexible interconnection, long-distance transmission

"Load"



Demand response, adjustable load, vehicle-to-grid interaction

"Storage"



Peak shaving and valley filling, smooth out fluctuations, and provide backup support

NEW POWER SYSTEM INTEGRATION & APPLICATION

GENERATION SIDE

Large-Scale Wind–Solar Bases



Desert



Gobi



Barren Land



Saline-Alkali Land

Abundant Wind & Solar Resources
Lower Land-Use Constraints
Concentrated Development Conditions
Suitable for Long-Distance Bulk Power
Delivery

2021 | Economy-Wide Strategic Priority

Desert–Gobi–Barren Land Bases Prioritized

2022 | Base-Oriented Development

14th Five-Year-Plan clarified centralized renewable base development

2024 | Rapid Commissioning

First batch of large bases accelerated into operation

2025 | Export + Multi-Energy Coordination

Large bases shift toward integrated delivery and coordination

Ningxia Tengger Desert New Energy Base

PV + Desert Restoration + Energy Storage + UHV Transmission

- ❑ 13 GW planned capacity
- ❑ 8.5 GW developed by Longyuan Power
- ❑ First 3 GW of PV commissioned
295 MW / 590 MWh energy storage
- ❑ 5.78 TWh annual generation



4.2million shrubs | 11 drought-tolerant species | Vegetation coverage: <0.5%→20%

NEW POWER SYSTEM INTEGRATION & APPLICATION

GRID SIDE

VSC-HVDC Transmission Technology



Long-Distance
Offshore Wind
Transmission



Renewable Energy
Base Grid
Integration



Power Supply to
Urban Load
Centers



Island Power
Supply

Offshore Wind Export

the voltage level ± 400 kV $\rightarrow$ ± 500 kV
transmission capacity 1.1–2.0 GW

Load-Center Interconnection

Greater Bay Area DC back-to-back project

UHV Flexible DC

Gansu–Zhejiang and Xizang–Guangdong DC projects, the voltage level ± 800 kV

UHVDC Transmission Technology



West-to-East
Power
Transmission



Power Export from
Desert–Gobi
Renewable Bases



Southwest
Hydropower
Transmission



Interprovincial &
Interregional
Backbone Corridors

Southwest Clean Energy Export

Yazhong–Jiangxi · Baihetan–Jiangsu · Baihetan–Zhejiang · Jinshang–Hubei

Desert–Gobi Renewable Base Export

Hami–Chongqing · Ningxia–Hunan · Longdong–Shandong · Northern Shaanxi–Anhui · Gansu–Zhejiang

Backbone Transmission Corridors

± 800 kV · 8 GW capacity · 36+ TWh annual delivery

NEW POWER SYSTEM INTEGRATION & APPLICATION

LOAD SIDE

Flexible Loads

	Type	Response Logic	Typical Resources	
	Interruptible Load	Temporary reduction or shutdown during peak hours	Industrial lines, steel, chemical industry, cement	
	Shiftable Load	Shift demand from peak to off-peak periods	Electric boilers, cold storage, wastewater treatment, selected industrial processes	
	Time-Shiftable Load	Same total energy use, different operating time	Laundry, refrigeration, heat pumps, EV charging	
	Curtailable Load	Short-term reduction of non-critical demand	Building HVAC, lighting, commercial loads	
	Upward Flexible Load	Increase demand during high renewable output	Electrolyzers, electric boilers, battery charging, flexible data center tasks	
	Bidirectional Flexible Load	Charge from and discharge to the grid	EVs with V2G, energy storage, microgrids	



NEW POWER SYSTEM INTEGRATION & APPLICATION

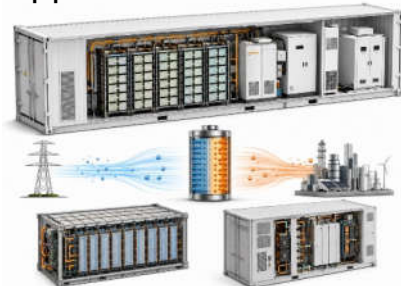
ENERGY STORAGE SIDE

Short-Duration Energy Storage for Frequency & Peak Regulation

Frequency Regulation · Inertia Support · Renewable Smoothing · Peak Shaving · Reserve · Black Start



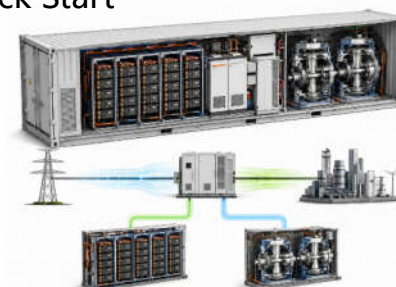
Li-ion Battery Storage



Sodium-ion Battery Storage



Flywheel Storage



Li-ion + Flywheel Hybrid Storage

Long-Duration Energy Storage

Peak Shifting · Daily/Weekly Balancing · Renewable Integration · Capacity Support · Seasonal Balancing



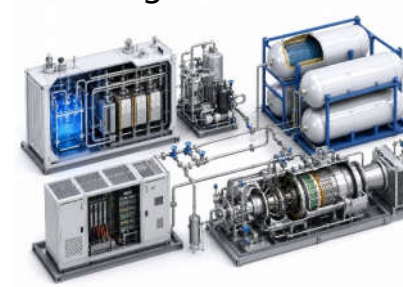
Flow Battery Storage



Compressed Air Energy Storage



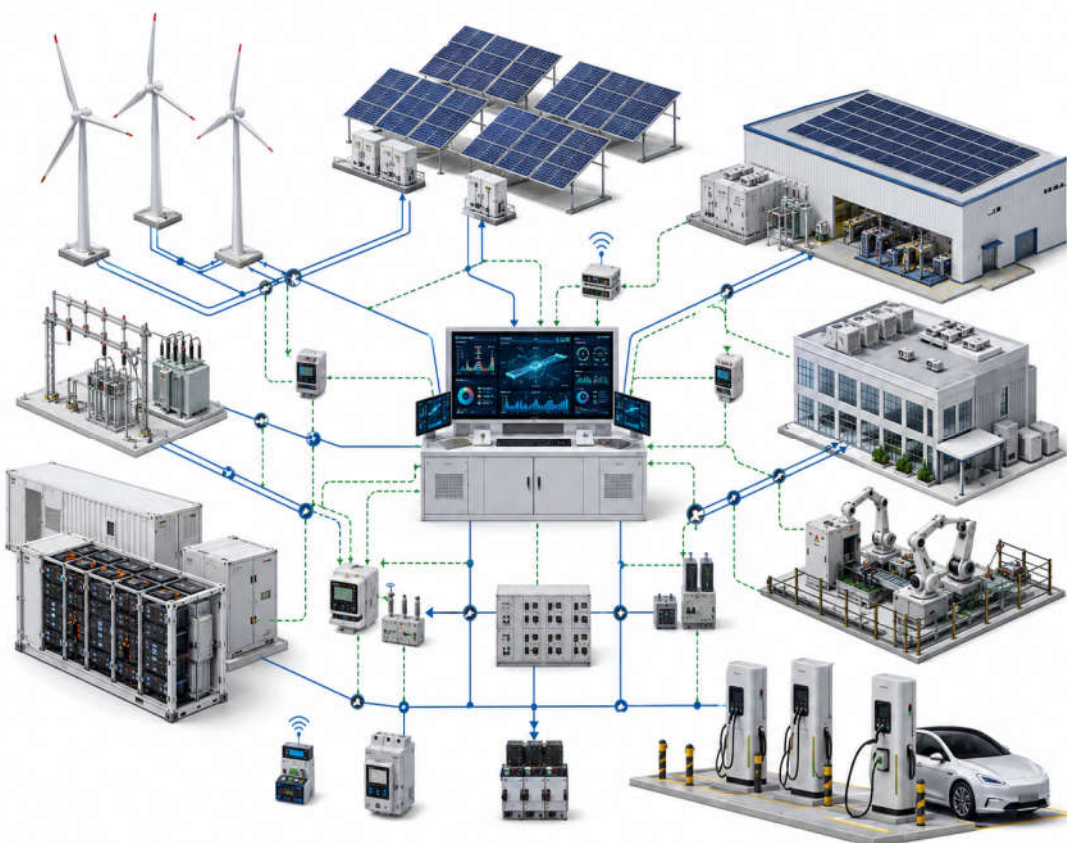
Gravity Energy Storage



Hydrogen Storage

NEW POWER SYSTEM INTEGRATION & APPLICATION

Smart Microgrid



Goldwind Beijing Yizhuang Carbon-Neutral Smart Park

1.8 MW distributed PV | 4.8 MW wind | 2.9 MWh storage | 3,000 RT·h chilled-water storage | 0.7 MW microturbine

CHN Energy Yantai Longyuan Low-Carbon Smart Park

2 MW rooftop PV | 600 kW carport PV | 3 kW distributed wind | 500 kW/1 MWh energy storage | IoT | big data & cloud EMS

Jinko Technology Shangrao Source-Grid-Load-Storage Microgrid

~5.99 MW rooftop PV | 0.9 MW / 1.8 MWh energy storage | 5G smart control & virtual power plant integration

Taolian Village Flexible Grid-Forming Smart Microgrid

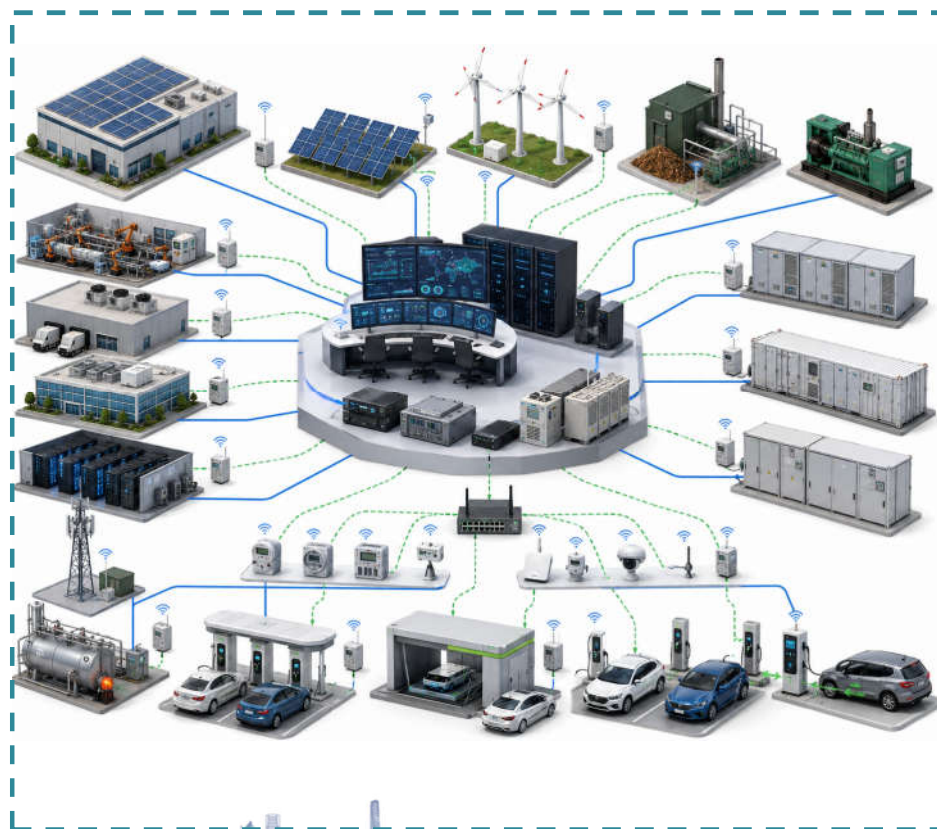
7 energy storage units | 8 small hydropower stations | 64 distributed PV sites | second-level fault switching

New Power System Capability Enhancement Pilots

Jilin Oilfield xinbei smart microgrid | Xuzhou Datun coordinated microgrid | Aheqi smart microgrid in Xinjiang

NEW POWER SYSTEM INTEGRATION & APPLICATION

Virtual Power Plant



Exploration Phase

Started with demand response and commercial building load management

Dec. 2019: Northern Hebei VPP commissioned

Platform Development Phase

From single-site pilots to city-level platform construction

2022: Shenzhen VPP Management Center established

Market Participation Phase

VPPs entered spot, ancillary service, and demand response markets

Sep. 2023: Shanxi launched VPP market trading

2024: Shandong registered its first 8 VPPs

Standardization & Scaling Phase

Unified definition, functions, market rules, and safety requirements

By 2027: VPP flexibility capacity > **20 GW**

By 2030: VPP flexibility capacity > **50 GW**

NEW POWER SYSTEM INTEGRATION & APPLICATION

Green Power Direct Connection

Dedicated Lines

Clear Traceability

Load-Matched Supply

Source-Load-Storage Coordination

CHN Energy-Donghua Steel 200 MW Wind Direct Supply

200 MW wind power | grid-connected | steelworks load

CHN Energy Yongzhou-Hunan Zhongtai Green Power Direct

Wind-solar-storage | dedicated line | polyimide film plant

Sun Metals Zinc Refinery Solar Direct Supply, AUS

124 MW solar | behind-the-meter | Connect to the 33/132 kV substation | meet 1/3 of the electricity consumption



Renewables supply designated users through dedicated lines instead of connecting directly to the public grid. Projects may operate on- or off-grid.



NEW POWER SYSTEM INTEGRATION & APPLICATION

Computing–Power Coordination

CHD Horinger Green Data Center Cluster

300 MW wind + 60 MW solar | dedicated supply |
64.8 MW / 259.2 MWh energy storage

CDT Zhongwei Green Computing Project

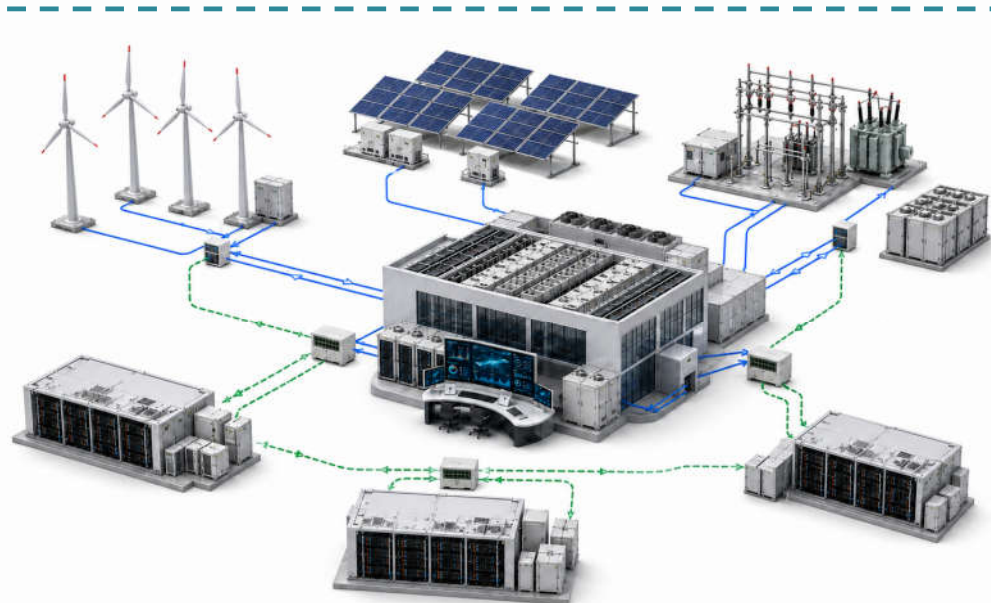
500 MW solar | dedicated-line supply | bilateral power
trading | storage-supported | commissioned May 2026

Malaysia Johor Green Data Center Cluster

YTL area: 500 MW planned
PDG JH1: The first phase of
52 MW has been delivered
AirTrunk: 150 MW has been
put into operation.
+ >270 MW planned
CRESS green power + energy
storage

Thailand EEC Computing Power Center Cluster

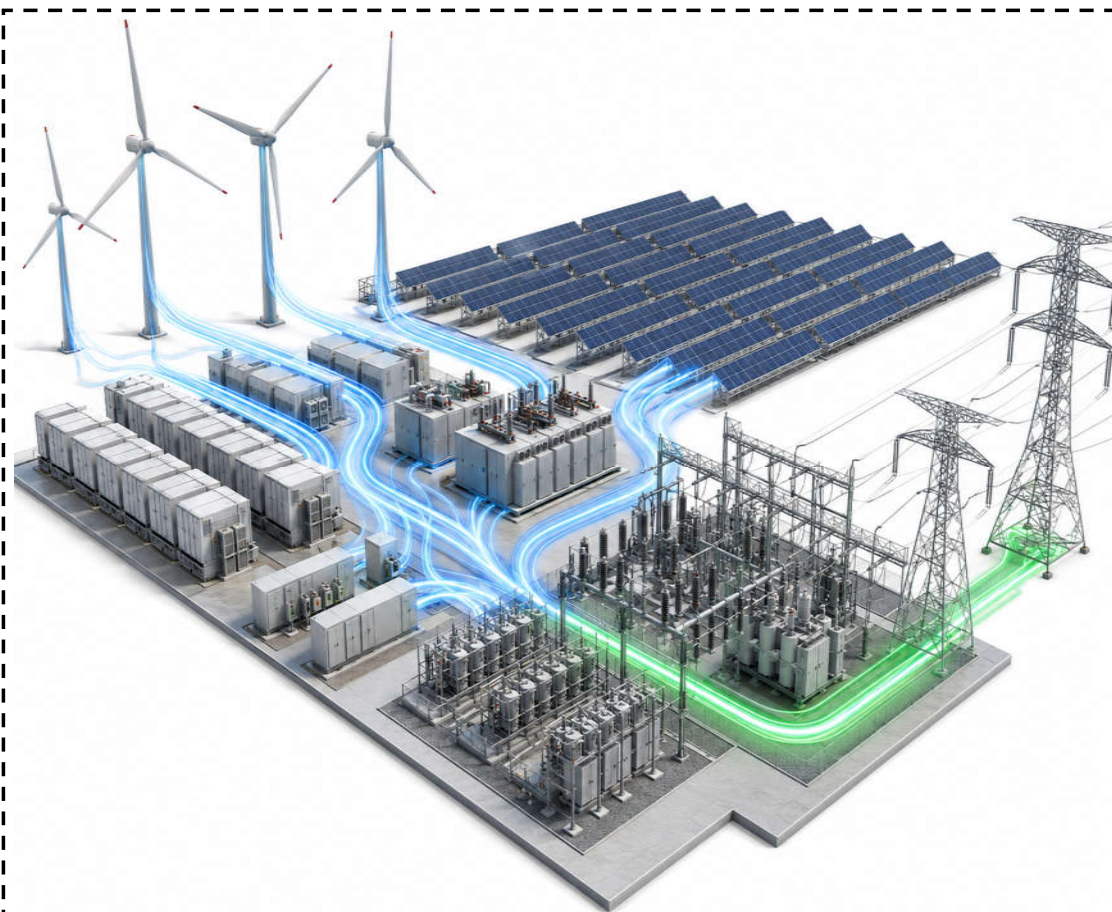
Google Chonburi invest: US\$1
billion
536 MW IT load capacity
approved in 2025
Skyline 200 MW + Bridge 134
MW approved in 2026
UGT2 green power mechanism
2GW Direct PPA pilot in progress



Computing–power coordination jointly optimizes power
supply, energy storage and workload scheduling, shifting
flexible computing toward cleaner and lower-cost electricity

NEW POWER SYSTEM INTEGRATION & APPLICATION

System-Friendly Renewable Energy Station



Coordinated Operation of Renewables & Energy Storage
Firm power output
Power supply assurance
Dispatchable reliable capacity

CTG Wulanchabu Grid-Friendly Green Power Station Demonstration Project

Energy storage is 30% of the installed capacity of wind & PV
Participate in peak load regulation, smooth fluctuations, power tracking, optimization operation of the spot market, and sharing regulation of new energy

The First Batch of NEA New Power System Pilots

Huainan, Anhui: Wind-PV-solar thermal-energy storage integration
Fuzhou, Fujian: Large-scale offshore wind power
Shannan, Tibet: Cold and high-altitude power system
Wulan, Haixi, Qinghai: Coordinated wind-PV-energy storage control

NEW POWER SYSTEM INTEGRATION & APPLICATION

Wind-Solar-Storage-Charging Integrated Stations

- ❑ **Shuangjie Hefei Integrated Energy Park:** 5.9 MWp PV, 400 kW wind, 250 kW/500 kWh energy storage, 240 kWp PV carport, 50 Charging parking space
- ❑ **Yinchuan Smart Charging Demonstration Center:** 2 sets of 10kW wind, 5 sets of 200 kW PV carports, 250 kW/500 kWh energy storage, serves 49 EV
- ❑ **Zaozhuang User-side Energy Storage Project:** 480 kW/1.72 MWh storage, 39.96 kWp PV carport, 10 kW wind, 250 kW charging capacity
- ❑ **Dawu Hongfan Village Integrated Station**
- ❑ **Ningdu Rural Microgrid Demonstration Project**
- ❑ **Lvliang Smart Energy Project:** 5 MWp Photovoltaic, 50 kW wind power, 5 MW / 20 MWh energy storage, 10 sets of 60 kW DC charging piles & 10 AC

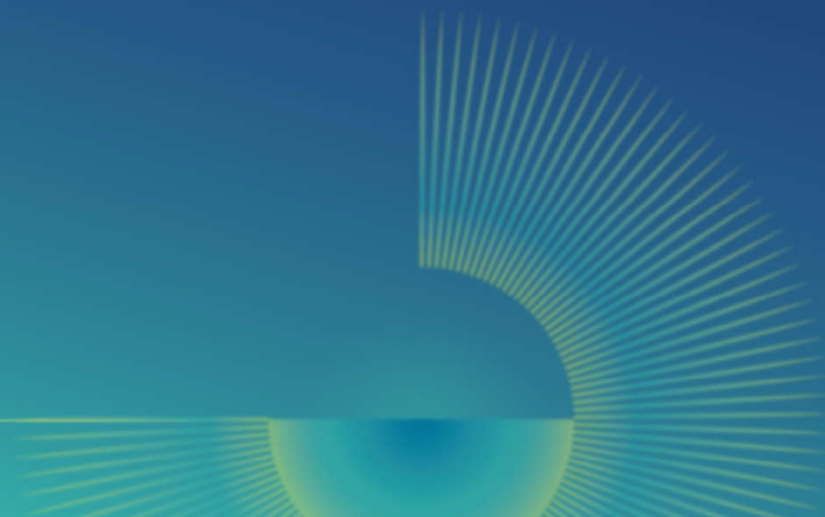
Vehicle-to-Grid Demonstration Zones

- ❑ **Beijing V2G Coordinated Control Pilot:** 3rd-generation bidirectional battery-swap station put into operation
- ❑ **Shanghai City-level V2G Pilot Projects:** 4 pilot projects, 7 types of aggregated resources, participate in bidirectional charging & discharging
- ❑ **Changzhou Distributed V2G Pilot:** 390 EVs, 9 charging stations, 15 MW flexible load
- ❑ **Shenzhen-Hong Kong V2G Pilot Project:** 151 V2G interaction stations, 840 MW adjustable load
- ❑ **Guangzhou City-level V2G Demonstration Project:** 430 MW V2G interaction aggregates resources, 1,000+ V2G chargers, 1,000+ V2G vehicles



FOUR

POLICY & MARKET FOR NEW ENERGY POWER GENERATION



RENEWABLE ENERGY POWER POLICY AND MARKETS

Dual Carbon Goals

14th Five-Year Renewable Energy Development Plan

Energy Law



Evolution of the New Energy Development Stage

RENEWABLE ENERGY POWER POLICY AND MARKETS

New energy development policies

2021	 《Carbon Peak Action Plan》	Scale up wind & solar (both centralized & distributed) 2030 Target: >1,200 GW capacity
2022	 《 14th Five-Year Renewable Plan 》	Shift to market-driven, grid-parity growth Build new supply model: "Renewable base + Coal support + UHV transmission"
2025	 《 Distributed PV & Green Power Direct Supply 》	Enable direct green power supply via dedicated lines Cross-sector integration: Combine with manufacturing, green hydrogen, low-carbon industrial clusters, and data centers
2026	 《 Multi-user Green Power Direct Supply 》	Expand direct supply to multiple users Promote park-level integration (source-grid-load-storage)

Market Outcomes

By Jul 2024



wind + solar capacity reached **1,206 GW** - about 6 years ahead of the 2030 target

By end-2024



total renewable capacity reached **1,889 GW** - about 56% of total installed power capacity

2025 outlook



renewable capacity may reach **2,340 GW** — about 60% of total capacity



RENEWABLE ENERGY POWER POLICY AND MARKETS

New energy consumption-related policies

2021



《Carbon Peak Action Plan》

Renewable consumption guarantee

2024



《New Energy Consumption Support Notice》
《2024 Renewable Power Consumption Responsibility Weights》

Supporting grid expansion
Faster grid-connection process
Provincial responsibility weights
Key-industry green power monitoring

2025



《2025 Renewable Power Consumption Responsibility Weights》
《Guiding Opinions on New Energy Consumption and Regulation》

Binding consumption targets
Multi-level regulation system
Centralized + distributed development
Integrated consumption assessment

2024 Outcome

Renewable energy utilization rate remained above **95%** nationwide

2024 Policy Focus

Consumption focus shifts to planning, grid connection, distribution capacity, and flexible resources

Responsibility Mechanism

Consumption responsibility extends new energy governance to the demand side

2030 Target

Build a coordinated multi-level consumption regulation system

Future Demand

Support reasonable consumption of over 200 GW of new renewable capacity added annually



RENEWABLE ENERGY POWER POLICY AND MARKETS

New policies related to the new power system

2023



《New Power System Development Blue Book》

2024



《Guiding Opinions on Strengthening Peak-Shaving Storage and Smart Dispatch Capacity》
《Action Plan for Accelerating the New Power System (2024–2027)》

2025



《Guiding Opinions on Accelerating Virtual Power Plant Development》
《Special Action Plan for Large-Scale New Energy Storage Deployment (2025–2027)》
《Guiding Opinions on Promoting High-Quality Grid Development》

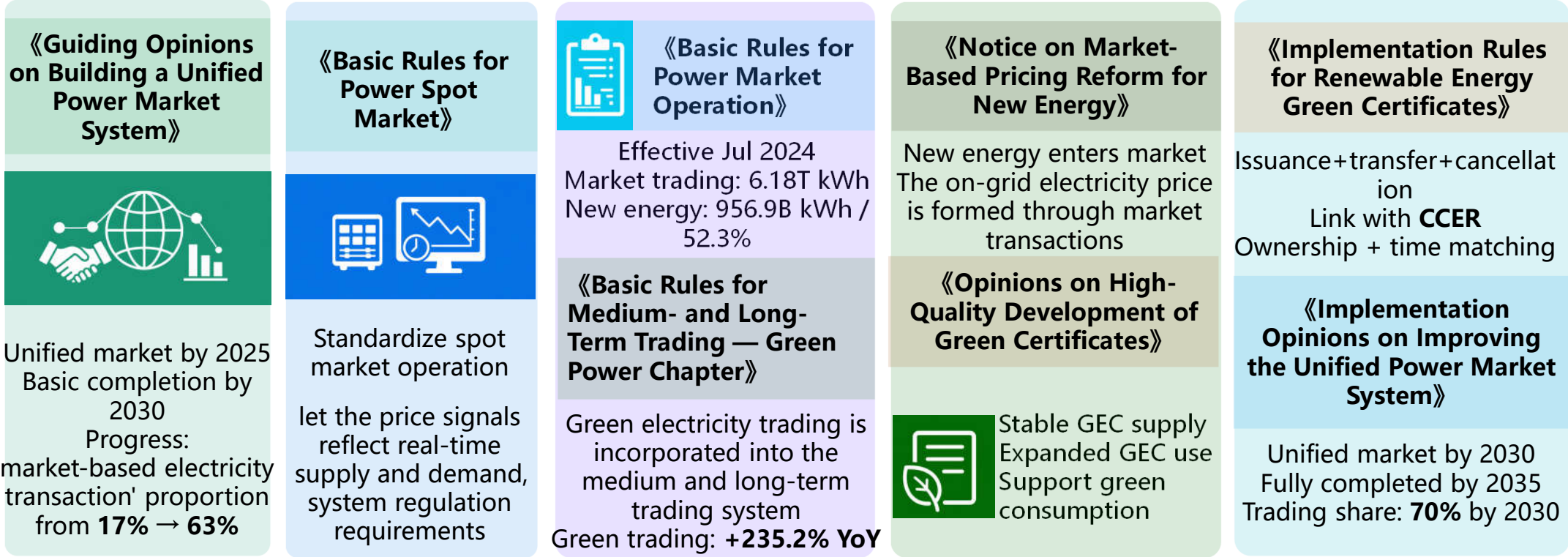
2030: accelerated transition stage 2030-2045: overall formation stage 2045-2060: consolidation and optimization stage	>80 GW pumped storage by 2027 Demand response above 5% of peak load Support >20% renewable generation share	20 GW+ VPP flexibility by 2027 180 GW+ new energy storage by end-2027 Promote high-quality grid development
--	--	---

New Power System Evolution Outlook

2023	2024	2027	2030
Demand response and integrated energy services growing	Grid-forming RE, storage, V2G, smart microgrids	VPP 20 GW+ Storage 180 GW+ Investment ~RMB 250 billion	Flexibility >5% Transmission 420 GW+ RE share in power generation ~30% Hosting distributed new energy 900 GW charging infrastructure numbers exceeds 40 million

RENEWABLE ENERGY POWER POLICY AND MARKETS

Electricity market-related policies



RENEWABLE ENERGY POWER POLICY AND MARKETS

Electricity market mechanism

Three-Tier Structure of the Market System



Ancillary Services Market

Frequency Regulation / Peak Shaving / Reserve / Ramping

Reflects the value of flexible regulation



Spot Market

Day-ahead / Intra-day / Real-time

Reflects real-time supply and demand and price fluctuations



Medium and Long-term Market

Annual / Monthly / Multi-day / Green Power PPA

Locks the electricity volume and price

Revenue Composition of New Energy Projects

New Energy Project Gains =



+



Medium and Long-term Locked Revenue

+



Spot Market Floating Revenue

+



Ancillary Services Compensation

+



Green Power / Green Certificate Value

+



Capacity Compensation or Mechanism Tariff Support

Core Logic

- Medium and Long-term: Stabilizes base revenue
- Spot: Reflects time-block value and fluctuation risks
- Ancillary Services: Reflects the value of regulation capability

RENEWABLE ENERGY POWER POLICY AND MARKETS

Green electricity & green certificate market

Policy-Driven

Market-Driven

Revenue-Driven for Renewables

1 New Energy Electricity

Wind, solar, biomass, hydro, etc.
Renewable power generation



2 Green Power Trading

Electricity revenue + green value,
Separate power and GEC pricing



3 GEC Issuance & Trading

1 GEC = 1,000 kWh renewable electricity,
The environmental attributes are traceable, tradable and redeemable



4 Corporate Green Consumption

Meet consumption responsibility targets
Raise green power use in key industries
Support green supply chains and ESG disclosure



5 Carbon Footprint / Carbon Labels / Export Competitiveness

For product carbon footprint, green power consumption certification, low-carbon product claims, international green trade



RENEWABLE ENERGY POWER POLICY AND MARKETS

Green electricity & green certificate market

Multiple Values of GECs

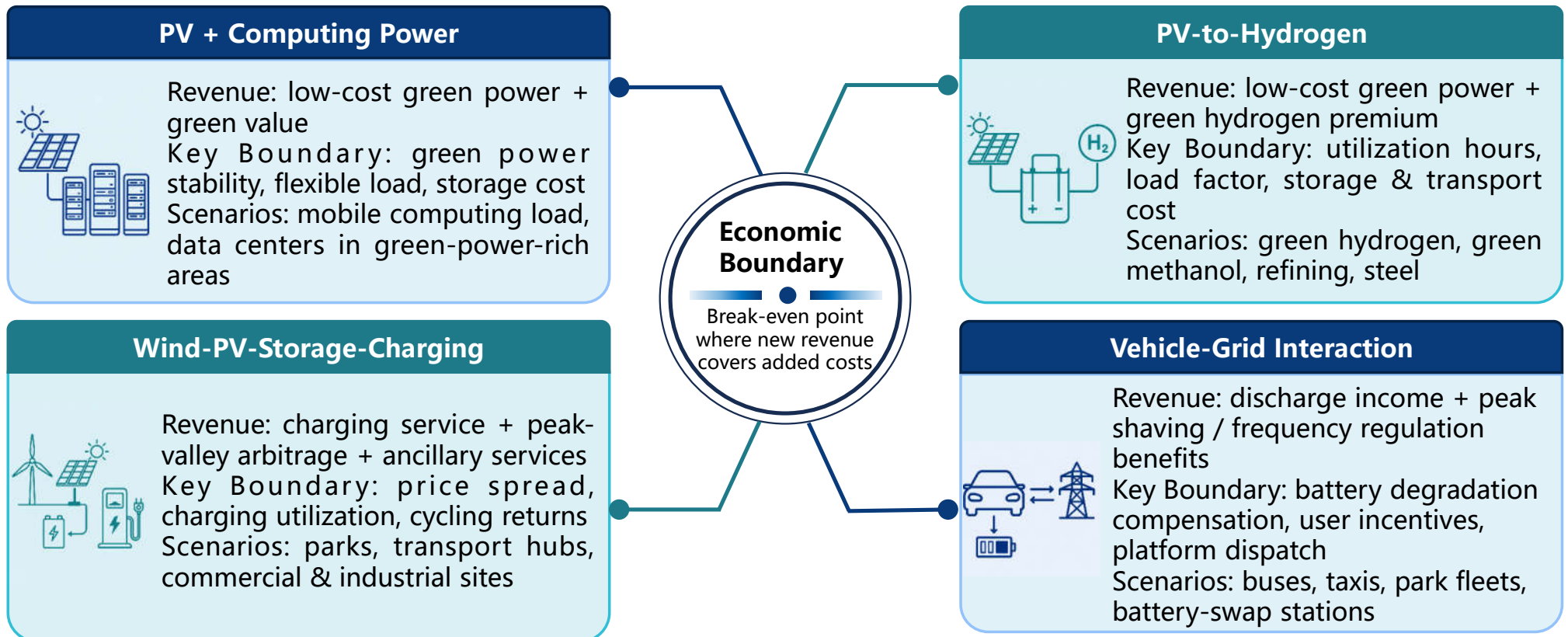
 Green Power Claim & Certification Support Green power use certification, RE100, zero-carbon factory claims	 Product Carbon Footprint & Labels basis for calculating indirect emissions of electricity, supports the declaration of product carbon footprint/ label	 Export Low-Carbon Certification & Competitiveness Meet green trade rules Improve export competitiveness	 Carbon Accounting & ESG Disclosure For carbon accounting ESG reporting Green procurement & supply chain management	 Policy Compliance & Assessment Consumption responsibility compliance Key-industry green power assessment
--	--	---	--	---

	Green Power Trading Volume	GEC Issuance Volume	GEC Trading Volume	Cumulative GEC Trading	GEC Measurement Standard
2025 Market Data	 328.5B kWh +38.3% YoY, 18 times larger than the scale of 2022 level	 2.947 billion GECs issued Including 1.893 billion tradable GECs	 930M GECs traded Growth rate increased by 1.08 times YoY Green certificates traded separately: 680 million Green electricity trading green certificates: 250 million	 1.483 billion GECs By end-2025	 1 GEC = 1,000 kWh renewable electricity

RENEWABLE ENERGY POWER POLICY AND MARKETS

Economic Boundary

Economic Boundary = Low-Cost Green Power + Flexible Load / Local Consumption + Market Revenue + Cost Control





CLEAN IN ACTION
GREEN
FOR
APEC

THANKS

